

```
import cv2
import numpy as np

#cap = cv2.VideoCapture(0) # initialize the camera
cap = cv2.VideoCapture("nvarguscamerasrc ! video/x-raw(memory:NVMM),format=NV12,

while True:
    _, frame = cap.read() # read a frame from the camera
    frame = cv2.resize(frame, (640, 480)) # resize the frame

    # apply color thresholding to detect white and yellow lanes
    hsv = cv2.cvtColor(frame, cv2.COLOR_BGR2HSV)
    lower_white = np.array([0, 0, 200])
    upper_white = np.array([180, 30, 255])
    mask_white = cv2.inRange(hsv, lower_white, upper_white)
    lower_yellow = np.array([15, 50, 50])
    upper_yellow = np.array([30, 255, 255])
    mask_yellow = cv2.inRange(hsv, lower_yellow, upper_yellow)
    mask = cv2.bitwise_or(mask_white, mask_yellow)

    # apply Gaussian blur to reduce noise
    blur = cv2.GaussianBlur(mask, (5, 5), 0)
```

Fig. 3: Code snippet for camera capture and image processing

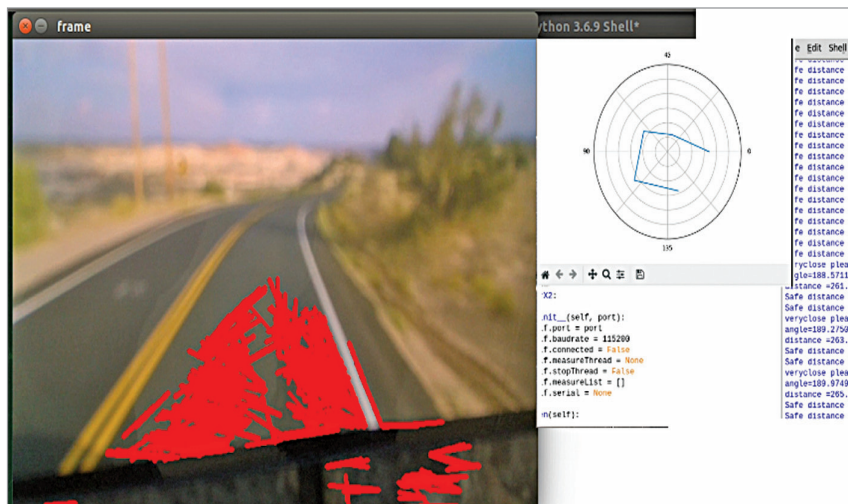


Fig. 4: Final testing lidar and lane detection of the ADAS system

check the following link for the options to power the kit:

<https://forums.developer.nvidia.com/t/power-supply-considerations-for-jetson-nano-developer-kit/71637>

To connect the camera to NVIDIA Jetson kit, you can either use a ribbon cable-based camera serial interface (CSI) camera or any USB camera. Connect the NVIDIA

Jetson kit's HDMI display and power the display through a USB. For the lidar's connection, please refer Part 1 of this article published in June issue.

Software

Prepare the NVIDIA Jetson kit with OS and Python using the instructions given in the following link:

<https://developer.nvidia.com/embedded/learn/get-started-jetson-nano-devkit>

Install the OpenCV computer vision library for processing the image needed in the ADAS system using Linux terminal and then install the NumPy Python library using follow-

ing commands:

```
sudo pip3 install opencv
sudo pip3 install numpy
```

After the above installations, you are ready to code. So, first import the NumPy and OpenCV and then set the camera for making the video frame using OpenCV. Next, create the loop function for processing the video frames and detecting the edges and pedestrians and cars on a road. The processed frame is shown in a loop on the HDMI display. Fig. 3 shows a snippet of the source code for the camera capture and image processing.

Combining anti-collision and lane detection systems

From Part 1 of this article, we already have the code to map surroundings with lidar and view the map in real time to alert when anything comes too close to the vehicle, to avoid accidents. Running both the codes (from Part 1 and Part 2) simultaneously shows the lidar map and the lane detection video on HDMI display that is attached to a vehicle's dashboard.

To do that, use the Python subprocess (see below) and create another code that runs both the Python scripts together. Now, after connecting the lidar with camera and powering the NVIDIA Jetson kit, you may run the ADAS Python script and see the output on the HDMI screen. Fig. 4 shows the final testing of the lidar and lane detection of the ADAS system.

```
import subprocess
subprocess.run("python3 script1.py &
python3 script2.py", shell=True)
```

Bonus. You can watch the video of the tutorial of this DIY project at <https://www.electronicsforu.com/electronics-projects/lidar-based-anti-collision-system> **EFY**

Ashwini Kumar Sinha, an IoT and AI enthusiast, is a tech journalist at EFY

EFY Note
The source code
of this project
is available
for free download at
www.electronicsforu.com